## TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

**Reminders**

**Intercepts:** set y = 0 to find the x-intercept; set x = 0 to find the y-intercept.

**Solution of a system:** the point where the lines cross. It makes both equations true.

**How many?** Different slopes: one. Same slope, different y-intercept: none. Same line: infinite.

**1.** What is the $x$-intercept of $2x + 5y = 20$?

**A.** (0, 4)

**B.** (4, 0)

**C.** (20, 0)

**D.** (10, 0)

**2.** What is the $y$-intercept of $3x + 4y = 24$?

**F.** (0, 6)

**G.** (8, 0)

**H.** (0, 8)

**J.** (6, 0)

**3.** Is (2, 3) a solution of $y = 2x - 1$?

**A.** No, because $3 \neq 5$

**B.** Yes, because $3 = 3$

**C.** Yes, because $2 = 2$

**D.** No, because $3 \neq 1$

**4.** What is the slope of $y = -\frac{1}{2}x + 4$?

**F.** 4

**G.** $\frac{1}{2}$

**H.** $-\frac{1}{2}$

**J.** $-2$